Class 8 Notes

Quick revision: Maths, English, Science

Maths

- **Rational numbers:** Closed under addition, subtraction, multiplication (not division by 0). Commutative and associative for addition and multiplication. Additive identity 0, multiplicative identity 1. Reciprocal of a/b is b/a.
- **Linear equations:** One variable: ax + b = c. Transpose terms and solve. 2x + 3 = 11, so x = 4. Check by substituting.
- **Quadrilaterals:** Angle sum = 360°. Polygon with n sides: sum of interior angles = (n - 2) × 180°. Sum of exterior angles = 360°. Parallelogram: opposite sides and angles equal, diagonals bisect each other. Rectangle: diagonals equal. Rhombus: diagonals bisect at right angles.
- **Squares and cubes:** Square root undoes squaring: √144 = 12. Cube root undoes cubing: $3\sqrt{27}$ = 3. Perfect square ends in 0, 1, 4, 5, 6 or 9.
- **Comparing quantities:** Discount = MP - SP, Discount % = (Discount / MP) × 100. Compound interest: $A = P\,(1 + R/100)^n$, CI = A - P. GST is a tax added to the price.
- **Algebraic identities:** $(a + b)^2 = a^2 + 2ab + b^2$. $(a - b)^2 = a^2 - 2ab + b^2$. $a^2 - b^2 = (a + b)(a - b)$. $(x + a)(x + b) = x^2 + (a + b)x + ab$.
- **Mensuration:** Trapezium area = 1/2 (a + b) h. Cuboid: volume lbh, total surface area 2(lb + bh + hl). Cube: volume a^3, TSA $6a^2$. Cylinder: volume $\pi r^2 h$, CSA $2\pi rh$, TSA $2\pi r(r + h)$.
- **Exponents:** $a^{-n} = 1/a^n$. $a^m \times a^n = a^{m+n}$. $(ab)^n = a^n b^n$. Standard form: 3.5×10^5.
- **Proportion:** Direct: x and y increase together, x / y is constant. Inverse: as x increases, y decreases, x × y is constant.
- **Factorisation:** Take out the common factor, group terms, or use the identities. $x^2 - 9 = (x + 3)(x - 3)$.
- **Data handling:** Histogram (class intervals, no gaps), pie chart (angle = value / total × 360°). Probability = favourable outcomes / total outcomes.

English

- **Reported speech:** Tense moves back: am/is → was, will → would, can → could, present perfect → past perfect. Questions: She asked if/whether ... Commands: He told me to ...
- **Active and passive:** Passive with each tense: is written, was written, has been written, will be written, is being written.
- **Conditionals:** Zero: If you heat ice, it melts. First: If it rains, I will stay. Second: If I were rich, I would travel.
- **Clauses and connectors:** Noun, adjective and adverb clauses. Linkers: although, however, in spite of, as soon as, unless.
- **Phrasal verbs and idioms:** give up (stop), look after (care for), turn down (refuse), call off (cancel). Use idioms naturally in writing.
- **Determiners and modals:** Use few/a few, little/a little, some/any correctly. Modals: ought to, need to, had better.
- **Writing:** Formal letter, article, speech, notice, report. Structure: heading, introduction, body, conclusion.

Science

- **Crop production:** Kharif crops (June-October): rice, maize, soybean. Rabi crops (October-March): wheat, gram, pea, mustard. Steps: preparing soil, sowing, adding manure and fertilisers, irrigation, weeding, harvesting, storage.
- **Microorganisms:** Bacteria, fungi, protozoa, algae, viruses. Useful: curd, bread, antibiotics (penicillin), vaccines. Harmful: cholera, typhoid, malaria (female Anopheles mosquito).
- **Synthetic fibres, plastics:** Rayon, nylon, polyester, acrylic. Thermoplastics soften on heating (polythene, PVC). Thermosetting plastics do not (bakelite, melamine).
- **Metals and non-metals:** Metals: lustrous, malleable, ductile, good conductors, sonorous. Non-metals: dull, brittle, poor conductors. Metal oxides are basic; non-metal oxides are acidic.
- **Coal and petroleum:** Fossil fuels formed from dead plants and animals. Petroleum gives petrol, diesel, kerosene, LPG, bitumen. Natural resources must be conserved.
- **Combustion:** Needs fuel, oxygen and heat to reach ignition temperature. Calorific value is heat produced by 1 kg of fuel. Fire extinguishers cut off the oxygen supply.
- **Cell:** Basic unit of life. Cell membrane, cytoplasm, nucleus. Plant cells also have a cell wall, chloroplasts and a large vacuole. Robert Hooke first saw cells (1665).
- **Reproduction in animals:** Sexual reproduction: fertilisation (internal in humans, external in frogs). Oviparous animals lay eggs; viviparous animals give birth to young.
- **Adolescence:** Puberty (11 to 19 years) brings body changes controlled by hormones. Sex determination: XX = female, XY = male (decided by the father's chromosome).
- **Force, pressure, friction:** Force is a push or pull. Pressure = force / area. Friction opposes motion; rolling friction < sliding friction < static friction.
- **Sound:** Produced by vibrations. Frequency in hertz. Human audible range 20 Hz to 20,000 Hz. Loudness depends on amplitude; pitch on frequency.
- **Electric current effects:** Chemical effect: electrolysis and electroplating. Liquids that conduct (acid, salt water) are good conductors.
- **Light and eye:** Laws of reflection. Eye parts: cornea, iris, pupil, lens, retina, optic nerve. Braille helps the visually challenged.
- **Solar system:** Planets in order: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune. A light year is the distance light travels in one year.